

F28WP Web Programming Coursework Group Project Specification

Assignments Guidelines - (30% of total course marks)

Contact Information:

Edinburgh

Dr Mehran Sharghi m.sharghi@hw.ac.uk

Dr Mario Soflano m.soflano@hw.ac.uk

Dubai

Course Leader: Dr Md Azher Uddin m.uddin@hw.ac.uk

Kayvan Karim K.Karim@hw.ac.uk

1. Introduction

This is a group coursework. For this coursework, you are required to design and develop an interactive web application of your choice. The interactive web application aims to create a comprehensive, secure, and user-friendly platform that seamlessly integrates client-side and server-side technologies to deliver a rich user experience. By leveraging HTML for structure, CSS for design, JavaScript for interactivity, jQuery for document manipulation, JSON for data transfer, Node.js for backend functionality, MySQL for data storage, and Web Storage for local data persistence, the application aims to provide robust functionality, efficient data management, and enhanced security to protect against common web threats. This coursework consists of two parts: a proposal for a web application CW1.1 and the development of the proposed application CW1.2.

2. Objectives

- Designing and developing client-server web applications
- Frontend development using HTML, CSS, JavaScript, jQuery, React.
- Backend programming using NodeJS
- Using databases in web applications
- Using code repositories in practice
- Gaining teamwork experience

3. Preparation and general rules

Teams (Groups)

- Students will self-select and form their own teams.
- Each team will be made up of five (5) students.

Teamwork

- Each team member must take ownership and implement some part of the application.
- You must become knowledgeable about the work done by your team members.
- You must contribute to the overall code, analysis, design and evaluation of the web application. This is a teamwork project; you are all accountable for it.

Tools and Technology

- Suggestion: The web application should have both client-side and server-side. Use HTML for structure, CSS for design, JavaScript for interactivity, jQuery for document manipulation, JSON for data transfer, Node.js for backend functionality, MySQL for data storage, and Web Storage for local data persistence.
- You host your web application on your local machine.

Web Application Selection

- Each team will have their own web application (e.g., Flight Booking, Hotel

Reservation, Rent Car and so on). No team can share a web application with another. Sharing is considered collusion and carries all the relevant consequences.

- Use Canvas to form your team.
- Please do not enrol the group in the wrong campus.

Expectation of Web Application

- The Web Application you design should cover at least six functional requirements, including insert data into database, modify data in database, retrieving data from database and delete data from database through the web app.

4. Instructions and Tasks

- Your group will deliver your design (or Proposal) presentation (CW 1.1) during your allocated session in Week 4 (You will receive an announcement on Canvas for your presentation date and time). This presentation should be short, less than 5 minutes (4 minutes for the presentation and 1 minute for Q/A), so you must be concise. All members must attend the presentation and deliver some part of it. If any members do not attend the presentation, they will be awarded an 'ABS' and 0 marks for this section of the coursework. **Note that the Canvas submission for CW1.1 should be done before 15:00 Monday 29th September 2025 (17:00 for Dubai).** The Canvas submission should include your proposal presentation in PPT format.
- The proposal presentation (CW 1.1) should include a title page with the names of group members, an introduction to the application including its aim and scope, a Use Case diagram of the application, Functional and non-functional requirements of the application, Tools and Techniques to be used, and a project plan.
- Final product (CW 1.2) your group will deliver your final product (Web Application) with a presentation during your allocated session in Week 11 (You will receive an announcement on Canvas for your presentation date and time). This presentation should be approximately 8 minutes (6 minutes for demo presentation and 2 minutes for Q/A). All members must attend the presentation and deliver some part of it. If any members do not attend the presentation, they will be awarded an 'ABS' and 0 marks for this section of the coursework. **Note that the Canvas submission for CW 1.2 should be done before 15:00 Wednesday 19th November 2025 (17:00 for Dubai).** The Canvas submission should include your demo video (6 minutes maximum), the code of the web application. In the demo video you should present your web application. The presentation is expected to have four parts. 1) A quick introduction of the group and the web application. 2) The high-level architecture of the application. This could be a diagram illustrating the main components of the system and their interactions. 3) A demonstration of all the functionalities of the application by running them interactively. This should take most of the presentation time. 4) A brief overview of the code structure.
- Note that the group will receive 0 marks for the coursework if the Canvas submission is not done appropriately as well if the Canvas submission is not accessible.

- During your final demo presentation, your group should present your web application to one of the teaching team members. The presentation structure is similar to the demo video except that there will be a Q&A at the end. All the members of the group are required to participate in the presentation and Q&A. For your presentation your group can prepare presentation slides, but it is not mandatory.
- The students are only allowed to use the tools and libraries which are covered in the lecture materials.

5. Deliverables – Key Dates

Summary of Schedule

Sl. no.	Activity	Deadline
1	Group Formation	Week 2
2	CW 1.1: Proposal Presentation	Week 4
3	CW 1.2: Final Presentation, Demo and Code	Week 11

6. Responsibilities

- You only need to submit one presentation per team. Normally the team Reporter is in charge of submitting the report, but everyone in the team is responsible for ensuring the submission is done correctly.
- The submission deadline for the proposal presentation is **Monday 29th September 2025 at 15:00 (17:00 for Dubai)**. The Canvas submission should include your proposal presentation in PPT format.
- **The submission deadline for the final presentation and demo video is 15:00 Wednesday 19th November 2025 (17:00 for Dubai).**
- Late submissions will have a (-30%) penalty applied as per university policy. Plagiarism and/or collusion will not be tolerated, and a 0% mark will be applied.

7. Feedback

- For the coursework, students will receive feedback on Canvas within three teaching weeks from the presentation date.
- According to the university policy Feedback or mark will not be released during Exams and Assessment Diets in order not to impact on students who might have exams during the Exams and Assessment Diet. [Read More](#).

8. Avoiding Plagiarism and Other Forms of Academic Misconduct

This is a reminder that you need to complete your coursework on your own with your group partners. Here are some further points to take into consideration:

- The usage of AI-generated content (such as text from ChatGPT) is forbidden and might result in voiding the course, resitting the course, and up to exclusion from the university, depending on your case and as per our disciplinary guidelines.
- Failure to reference work that has been obtained from other sources or to copy the words

and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course.

- Students must never give hard or soft copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code. Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.
- And remember: the consequences of taking dishonest shortcuts in coursework are much worse than getting a bad mark (or even no marks) on a piece of coursework. There have been cases where a student was awarded an ordinary degree (rather than an Honours degree) because of the sanction imposed by the University's Discipline Committee due to plagiarism in coursework.
- Further information on academic misconduct can be found at [Student-Academic Misconduct-Policy](#).

Please contact Dr Mehran Sharghi m.sharghi@hw.ac.uk (for Edinburgh students) or Dr Md Azher Uddin M.Uddin@hw.ac.uk (for Dubai students) if you have any questions or need further guidance.

Marking Rubrics

Design Presentation = 5%

Each criterion – apart from <i>Presentation quality</i> – is marked along the following scale: 0 – Missing: the criterion is not included in the presentation 1 – Insufficient: the criterion is missing several elements, or severely lacking in quality 2 – Poor: the criterion is incomplete in a few aspects 3 – Good: the criterion is complete 4 – Excellent: claims/decisions are justified appropriately 5 – Outstanding: creativity/out-of-the box thinking is displayed in the analysis/design		
Criterion	Description	Marks
Aim and Scope	The aim and scope of the web application are clear.	/1
Use Case Diagram	Present the use case diagram of the web application	/1
Functional and Non-Functional requirements	Functional and Non-functional requirements are clearly demonstrated.	/1
Tools and Technique	Tools and Techniques to be used are presented.	/1
Presentation quality	The presentation is well organised; the progression is fluid; the presenters are clear; the presenters are engaging with the audience; the slides are polished.	/1

Final Demonstration = 25%

The 1 st item is marked along the following scale: 0 – Missing: the criterion is not included in the presentation (1 – 2) – Insufficient: the criterion is missing several elements, or severely lacking in quality (2 – 4) – Poor: the criterion is incomplete in a few aspects (4 – 6) – Good: the criterion is complete (6 – 8) – Excellent: claims/decisions are justified appropriately (9 – 10) – Outstanding: creativity/out-of-the-box thinking is displayed in the analysis/design/implementation		
Each criterion – apart from 1 st item – is marked along the following scale: 0 – Missing: the criterion is not included in the presentation 1 – Insufficient: the criterion is missing several elements, or severely lacking in quality 2 – Poor: the criterion is incomplete in a few aspects 3 – Good: the criterion is complete 4 – Excellent: claims/decisions are justified appropriately 5 – Outstanding: creativity/out-of-the box thinking is displayed in the analysis/design/implementation		
Criterion	Description	Marks
Overall Application	The web application is well functional.	/10
Use of Tools and Techniques	All the tools and techniques are used appropriately.	/5
Code structure and quality	The web application code architecture is presented, and the code is well organised/structured.	/5
Presentation quality	The presentation is well organised; the progression is fluid; the presenters are clear; the presenters are engaging with the audience.	/5